

Density

Answer key with worked solutions

A. The idea

1. Definition.

Density is the **mass per unit volume** of a substance: $\rho = \frac{m}{V}$. Common units: **g/cm³** (solids), **g/mL** (liquids) and the SI unit **kg/m³**.

B. Calculating density

2. Densities.

$$(a) \rho = \frac{96.0 \text{ g}}{12.0 \text{ cm}^3} = \mathbf{8.00 \text{ g/cm}^3}$$

(b) Convert first: 2.50 kg = 2500 g and 0.500 L = 500. mL.

$$\rho = \frac{2500 \text{ g}}{500. \text{ mL}} = \mathbf{5.00 \text{ g/mL}}$$

3. Metal cube.

$$(a) V = (3.00 \text{ cm})^3 = 27.0 \text{ cm}^3, \text{ so } \rho = \frac{241 \text{ g}}{27.0 \text{ cm}^3} = \mathbf{8.93 \text{ g/cm}^3}$$

(b) Closest value: **copper** (8.96 g/cm³).

(c)

$$\begin{aligned} \text{error} &= \frac{8.926 \text{ g/cm}^3 - 8.96 \text{ g/cm}^3}{8.96 \text{ g/cm}^3} \times 100\% \\ &= \mathbf{-0.4\%} \end{aligned}$$

4. Displacement.

$$V = 45.2 \text{ mL} - 30.0 \text{ mL} = \mathbf{15.2 \text{ mL}} = 15.2 \text{ cm}^3$$

$$\rho = \frac{68.4 \text{ g}}{15.2 \text{ cm}^3} = \mathbf{4.50 \text{ g/cm}^3}, \text{ closest to } \mathbf{\text{titanium}} \text{ (4.51 g/cm}^3\text{)}.$$

C. Using density

5. Layers.

- (a) From the top: **oil** (0.92 g/cm^3), **water** (0.998 g/cm^3), **honey** (1.42 g/cm^3) at the bottom.
(b) The bead (1.10 g/cm^3) is denser than water but less dense than honey, so it sinks through the oil and water and **rests on top of the honey layer**.

6. Mass and volume.

- (a) $m = \rho V = 0.789 \frac{\text{g}}{\text{mL}} \times 200. \text{mL} = 157.8 \text{ g} = \mathbf{158 \text{ g}}$
(b) $1.00 \text{ kg} = 1000 \text{ g}$, so $V = \frac{m}{\rho} = \frac{1000 \text{ g}}{11.3 \text{ g/cm}^3} = \mathbf{88.5 \text{ cm}^3}$

D. Replicate measurements

7. Five replicates.

(a) Mean:

$$\bar{\rho} = \frac{(2.68 + 2.71 + 2.69 + 2.72 + 2.70) \text{ g/cm}^3}{5} = \frac{13.50 \text{ g/cm}^3}{5} = 2.70 \text{ g/cm}^3$$

Deviations: -0.02 , $+0.01$, -0.01 , $+0.02$ and 0.00 g/cm^3 ; the sum of their squares is $1.0 \times 10^{-3} (\text{g/cm}^3)^2$.

$$s = \sqrt{\frac{1.0 \times 10^{-3} (\text{g/cm}^3)^2}{5 - 1}} \\ = 0.016 \text{ g/cm}^3$$

(b) $\mathbf{2.70 \pm 0.02 \text{ g/cm}^3}$ ($n = 5$)

(c) The mean equals the accepted value for aluminium (2.70 g/cm^3), so the result is **accurate**; the standard deviation is small (under 1 % of the mean), so it is **precise**.

E. Challenge

8. The crown.

(a) 1.50 kg = 1500 g and 93.0 mL = 93.0 cm³:

$$\begin{aligned}\rho &= \frac{1500 \text{ g}}{93.0 \text{ cm}^3} = \mathbf{16.1 \text{ g/cm}^3} \\ 16.1 \frac{\cancel{\text{g}}}{\cancel{\text{cm}^3}} &\times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}} \times \frac{10^6 \cancel{\text{cm}^3}}{1 \text{ m}^3} \\ &= \mathbf{1.61 \times 10^4 \text{ kg/m}^3}\end{aligned}$$

(b) **No.** Pure gold is 19.3 g/cm³. The crown's lower density shows it contains a less dense metal, such as silver or copper, mixed with the gold.